

Treatment of Symptomatic Patients With Essential Thrombocythemia: Effectiveness of Anagrelide

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We prospectively evaluated the effect of anagrelide (ANA) on platelets, PF4, F1+2, PAP, PAI-1, and TFPI and erythromelalgia in patients with essential thrombocythemia (ET) receiving anti-aggregants both pre- and post-ANA. At first, we observed a successful reduction of platelets, which was associated with normalization of platelet coagulant and endothelial function and disappearance of erythromelalgia. Secondly, we found a correlation between PF4 and TFPI and between TFPI and thrombosis, suggesting that erythromelalgia may be caused by platelet-mediated endothelial activation. These data may indicate that ANA may be efficacy in the treatment of symptomatic patients with ET. *Am. J. Hematol.* 80:81–83, 2005. © 2005 Wiley-Liss, Inc.

Key words: anagrelide; erythromelalgia; essential thrombocythemia

Erythromelalgia frequently complicates the essential thrombocythemia (ET) [1]. Thrombocytosis and platelet coagulant activation are risk factors for this microvascular complication [1,2]. Its management includes cyto-reduction and anti-aggregants [1]. It has been demonstrated that ET patients receiving cyto-reduction and anti-aggregants show normal platelet count but still present platelet coagulant activation and thrombosis [3]. Anagrelide (ANA) is a platelet-lowering drug that inhibits platelet aggregation [4]. Therefore, we evaluated the effect of ANA on platelets and markers of platelet coagulant activation, such as platelet factor 4 (PF4), prothrombin fragment 1 + 2 (F1 + 2), plasmin/ α_2 -antiplasmin complex (PAP), and plasminogen activator inhibitor-1 (PAI-1) in patients with ET and erythromelalgia. Platelet endothelial activation releases tissue factor pathway inhibitor (TFPI), which is a marker of microvascular damage [5]. Thus we also determined TFPI. All measurements were performed before ANA and when complete response (CR), defined as platelets $< 500 \times 10^9/L$ for more than 1 month, was achieved. The study comprised 17 patients [10 men, 7 women; mean age 48 years (range 24–72 years)] with ET diagnosed according to criteria defined by the Polycythemia Vera Study Group [6]. The mean duration of disease was 4 years

(range 1–14 years). All patients were on anti-aggregants, such as aspirin (ASA) (9 patients), indobufen (IND) (7 patients), or dipyridamole (DYP) (1 patient). Nine of 17 patients suffered from erythromelalgia. These patients typically presented with painful burning and red congested extremities [1]. After a median time from diagnosis of 3 years, all patients were started on ANA. ANA was initially administered at a dose of 0.5 mg/day. Subsequently, the dose was increased by 0.5 mg/day every week until the platelets had decreased to below $500 \times 10^9/L$. The average maintenance dose was 2.1 mg/day (range 0.5–6 mg/day). Therapy was well tolerated. None of the patients had acquired thrombotic or inherited risk factors. Twenty healthy subjects served as controls. Platelets were determined on the Sysmex XE-21300 (Dasit, Milan, Italy). PF4, F1 + 2, PAP, PAI-1, and TFPI were measured by enzyme-

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TABLE I. Bioclinical Data and Statistics of ET Patients*

Patients	Anti- aggregants	Pre-anagrelide						Post-anagrelide					
		Platelets ($\times 10^9/L$)	PF4 (IU/mL)	F1+2 (nmo/L)	PAP (ng/mL)	PAI-1 (ng/mL)	TFPI (ng/mL)	Platelets ($\times 10^9/L$)	PF4 (IU/mL)	F1+2 (nmo/L)	PAP (ng/mL)	PAI-1 (ng/mL)	TFPI (ng/mL)
1 (E)	ASA	825	104	4.3	295	200	56	341	10	0.2	218	40	50
2	IND	1,010	250	12.6	473	200	157	499	4.4	2.1	0	39	32
3(E)	ASA	814	83	2.4	340	63	223	456	10	1.7	0	30	174
4(E)	IND	1,120	114	4.9	539	54	279	405	4.8	2	99	25	131
5(E)	IND	858	105	4	491	99	200	338	10	0.1	0	22	120
6	IND	1,731	102	4.9	0	200	81	347	10	2.2	0	37	32
7(E)	IND	1,252	130	1.3	291	200	257	422	2.5	1.1	290	40	146
8(E)	ASA	969	101	2.5	344	200	151	319	10	0.4	267	39	100
9	IND	1,204	96	2.1	560	71	102	451	10	0.1	487	40	90
10	ASA	747	200	1.1	0	66	249	315	15	0.8	0	25	248
11(E)	ASA	1,000	100	2.5	487	98	148	512	10	1.3	389	40	49
12	ASA	1,043	106	1.6	383	82	183	456	10	2.9	218	40	88
13	ASA	1,585	106	8.3	389	200	88	417	10	1.6	358	35	86
14(E)	IND	1,302	105	2.5	511	95	247	346	5	1.3	487	40	56
15(E)	ASA	1,464	92	0.8	100	200	155	367	10	0.1	0	35	100
16	DYP	1,163	115	10	1078	200	170	333	4	0.6	414	36	81
17	ASA	1,013	100	1.3	500	91	81	380	10	0.2	0	32	89

*E, erythromelalgia. Normal ranges: platelets, $(150-450) \times 10^9/L$; PF4, 1–10 IU/mL; F1+2, 0.4–1.1 nmol/L; PAP, 0–514 ng/mL; PAI-1, 2–47 ng/mL; TFPI 75–120 ng/mL. *P* values: PF4/TFPI, 0.002; TFPI/thrombosis, 0.015.

linked immunosorbent assay (Diagnostic Stago, Boehringer Mannheim, Mannheim, Germany; Dade Behring Marburg GmbH, Marburg, Germany; American Diagnostica Inc., Greenwich, CT). As levels of these parameters may be a consequence of preparation or thrombosis, we repeated these measurements in a further sample, collected after 1 or 2 months, in 17 patients. Both pre- and post-ANA measurements showed concordance for studied parameter concentrations. Before ANA, all patients had high platelets ($[1.120 \pm 272] \times 10^9/L$) and higher PF4 and F1+2 (118 ± 42 IU/mL and 3.9 ± 3.3 nmol/L, respectively) levels than controls (4.9 ± 3.1 IU/mL and 0.6 ± 0.3 nmol/L, respectively) ($P < 0.0001$ and $P < 0.0001$, respectively) as well as increased PAP, PAI-1, and TFPI (398 ± 248 ng/mL and 136 ± 62 ng/mL and 166 ± 69 ng/mL, respectively) levels compared to controls (128 ± 77 , 26 ± 8 , and 81 ± 12 ng/mL, respectively) ($P < 0.0001$, $P < 0.0001$, and $P < 0.0001$, respectively). After ANA, all patients had platelets $< 500 \times 10^9/L$ ($[394 \pm 63] \times 10^9/L$) and normal PF4 (8.5 ± 3.2 IU/mL), F1+2 (1.3 ± 0.6 nmol/L), PAP (189 ± 189 ng/mL), PAI-1 (35 ± 6 ng/mL), and TFPI (98 ± 54 ng/mL). The erythromelalgic symptoms disappeared in all cases. A positive correlation was noted between PF4/F1+2, PF4/PAP, PF4/PAI-1 and PF4/TFPI ($P < 0.0001$ and $P = 0.045$ and $P = 0.039$ and $P = 0.002$, respectively). No correlation was found between platelets, PF4, F1+2, PAP, PAI-1, and

thrombosis, whereas a significant correlation was found between TFPI and thrombosis ($P = 0.015$) (Table I). These results could indicate that ANA may also induce normalization of platelet coagulant and endothelial function, which is accompanied by disappearance of erythromelalgic signs and symptoms. The normalization of platelet function associated with disappearance of erythromelalgia has also been reported by Laguna et al. in ET treated with ANA [7,8]. In these studies, the disappearance of spontaneous platelet aggregation and decreased levels of thromboxane B_2 emerged as markers of normalized platelet function associated with remission and relief of thrombotic manifestations. In our report, decreased PF4, F1+2, PAP, and PAI-1 emerged as markers of normalized platelet coagulant function associated with remission and relief of thrombotic manifestations. van Genderen et al. observed increased β -thromboglobulin and thrombomodulin in ET patients and erythromelalgia [9]. We found increased PF4 and TFPI in ET patients and erythromelalgia and a positive correlation between PF4 and TFPI and TFPI and thrombosis. These data suggest that erythromelalgia may be caused by platelet-mediated endothelial activation and that ANA may be efficacious in the treatment of symptomatic patients. Further prospective studies are needed to establish whether one or more of these parameters can be appreciated as an additional variable in clinical models that may explain the beneficial effects of ANA.

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